

6.4.4 Nuclear fission and fusion

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) Einstein's mass–energy equation; $\Delta E = \Delta mc^2$	HSW9, HSW10, HSW12 Decision making process when building new nuclear power stations. Learners will also require knowledge of 5.1.4
(b) energy released (or absorbed) in simple nuclear reactions	
(c) creation and annihilation of particle–antiparticle pairs	
(d) mass defect; binding energy; binding energy per nucleon	
(e) binding energy per nucleon against nucleon number curve; energy changes in reactions	
(f) binding energy of nuclei using $\Delta E = \Delta mc^2$ and masses of nuclei	
(g) induced nuclear fission; chain reaction	
(h) basic structure of a fission reactor; components – fuel rods, control rods and moderator	
(i) environmental impact of nuclear waste	
(j) nuclear fusion; fusion reactions and temperature	
(k) balancing nuclear transformation equations.	